Sahil Betal

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SUMMARY

Backend-focused Computer Science student experienced in building scalable APIs, distributed data systems, and security research tools. Skilled in Python, FastAPI, Flask, and cloud-native backend development with emphasis on performance, observability, and system reliability.

EDUCATION

Vishwakarma Institute of Technology *Aug 2023 – May 2027*
Bachelor of Technology in Computer Science *Pune, India*

TECHNICAL SKILLS

Languages: Python, C/C++, SQL, JavaScript
Backend Development: FastAPI, Flask, REST APIs, Node.js, Async Programming, API Design
Systems & DevOps: Linux, Docker, Kubernetes, Nginx, Git, CI/CD, Shell Scripting
Cloud & Data: AWS, GCP (Vertex AI, Cloud Run), MySQL, Pandas, NumPy, Plotly, Matplotlib, ETL Pipelines

SOFTWARE ENGINEERING PROJECTS

Hybrid Graph RAG Medical QA System *2025*
Python, Flask, Neo4j, Chroma, LLM Systems

- Engineered a hybrid Retrieval-Augmented Generation architecture combining vector search and knowledge graph retrieval for medical question answering.
- Integrated FDA DailyMed drug labels and MedlinePlus datasets to build a structured medical knowledge base with entity-level relationship mapping.
- Implemented parallel QA pipelines (LLM-only, Vector RAG, Hybrid Graph RAG) with automated evaluation to select the most accurate response.
- Built a Flask backend exposing REST APIs and an interactive dashboard for query comparison and performance metrics.
- Optimized graph traversal queries in Neo4j to reduce retrieval latency, improving end-to-end response time across multi-hop reasoning chains.

Custom Honeypot for Multi-Protocol Threat Analysis *2025*
Python, FastAPI, Networking, Systems Engineering

- Developed a lightweight multi-protocol honeypot emulating HTTP, SSH, FTP, and SMB services to monitor real-world malicious traffic.
- Captured and analyzed 150+ intrusion attempts originating from 30+ attacker IPs for behavioral threat analysis and attacker fingerprinting.
- Designed an event-driven logging pipeline and asynchronous network handlers to handle concurrent connections reliably under high load.
- Generated structured threat intelligence reports by correlating IP geolocation, attack vectors, and session payloads across captured sessions.

AURA — Data Visualization & Analytics Platform *2025*
Flask, Plotly, Backend Systems

- Developed a backend analytics platform enabling automated ingestion, statistical analysis, and visualization of CSV datasets.
- Implemented REST APIs for dataset processing and dynamic chart generation used for exploratory data analysis workflows.
- Improved dashboard responsiveness through batching and caching within the data processing pipeline, reducing repeated computation overhead.
- Designed a modular chart-generation engine supporting multiple Plotly visualization types, configurable at runtime via API parameters.

MediaPipe Face Landmarker *2024*
JavaScript, MediaPipe, Computer Vision

- Built a real-time facial landmark detection system enabling avatar animation using MediaPipe's face mesh solution.
- Integrated 52 facial blendshape parameters for accurate expression tracking and fine-grained avatar control.
- Implemented a low-latency rendering pipeline supporting 20+ animated avatars simultaneously with smooth frame synchronization.
- Optimized WebGL-based rendering to maintain consistent frame rates under multi-avatar load without degrading detection accuracy.

EXTRACURRICULARS & VOLUNTEERING

TEDxVITPune — Web Developer *2025 – Present*
Official TEDx Chapter, Vishwakarma Institute of Technology *Pune, India*

- Designed and developed the official TEDxVITPune 2026 website, handling end-to-end frontend development and deployment.
- Collaborated with the core organizing team to align the website's design language with TEDx brand guidelines and event requirements.

NGO Volunteer — Amazon Community Initiative *2024*
Environmental & Community Outreach Program

- Participated in Amazon-supported NGO volunteering activities focused on community welfare and environmental awareness.
- Contributed to on-ground outreach efforts alongside a cross-functional volunteer team to drive sustainable community impact.

ACHIEVEMENTS & RESEARCH

Presented *Facial Landmark Detection and Avatar Animation* at ICAIN 2025 (Springer Proceedings).

Research work *Beta-Carotene: Molecular Structure C40H56* currently under review in IJSRR (ISSN: 2279-543X).

Ranked 13th among 110+ teams at InnoHack for backend architecture of CodeDebugger & Run.

CERTIFICATIONS

Cloud Technology Consultant — Coursera

AWS Cloud Solutions Architect — Coursera

NVIDIA Deep Learning Certificate — NVIDIA

Machine Learning Operations (MLOps) for Generative AI — Google

Machine Learning Operations (MLOps) with Vertex AI: Model Evaluation — Google